Bahria University, Islamabad

Department of Software Engineering

Object Oriented Programming Lab (Spring-2026)

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Lab Journal: 01

Date: 04/03/2026

Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1					
2					
3					
4					
5					

Comments:

Signature

Lab Tasks

https://github.com/MuhammadMuaaz-03/Assign5.git

Task 01

Write a program by using a class Counter to create two objects c1 and c2 of that class. Use constructor to initialize the value of count. Display the initial count value in the objects.
Then use the overloaded ++ operator (overloaded incrementing unary operator) to increment c1 once and c2 twice and display the resulting values.

Code

```
#include<iostream>
using namespace std;
class Counter
{
private:
	int count;
public:
	Counter();
	Counter(int x);
	Counter operator++(int);
	int showData();
};
Counter::Counter() : count(0) {}
Counter::Counter(int x) : count(x) {}
Counter Counter ::operator++(int)
{
	count++;
	return count;
}
int Counter::showData()
{
	return count;
}
int main()
{
	Counter C1(5), C2(10);
	cout << "Initial value of C1=" << C1.showData() << endl;
	cout << "Initial value of C2=" << C2.showData() << endl;
	C1++;
	C2++;
	C2++;
	cout << "Value after incrementing in C1=" << C1.showData() << endl;
	cout << "Value after incrementing in C2=" << C2.showData() << endl;

}
```

Output (Screenshot)

```
Microsoft Visual Studio Debu
Initial value of C1=5
Initial value of C2=10
Value after incrementing in C1=6
Value after incrementing in C2=12

D:\Muaaz\University\SEM 2\OOP lab\Assign\Assign5\x64\Debug\Assign5.exe (process 178
To automatically close the console when debugging stops, enable Tools->Options->Deb
le when debugging stops.
Press any key to close this window . . .
```

Task 02

*Write a program to overload the arithmetic operations (+, -, *, /) using the concept of operator overloading.*

Code

```
#include<iostream>
using namespace std;
class Overloadcal
{
private:
	int num;
public:
	Overloadcal();
	void setData(int);
	int showData();
	Overloadcal operator+(Overloadcal);
	Overloadcal operator-(Overloadcal);
	Overloadcal operator*(Overloadcal);
	Overloadcal operator/(Overloadcal);
};
Overloadcal::Overloadcal() : num(0) {}
void Overloadcal:: setData(int c)
{
	num = c;
}
int Overloadcal::showData()
{
	return num;
}
Overloadcal Overloadcal:: operator+(Overloadcal n)
{
	Overloadcal d;
	d.num = num + n.num;
	return d;
}
Overloadcal Overloadcal::operator-(Overloadcal n)
{
```

```
        Overloadcal d;
        d.num = num - n.num;
        return d;
}
Overloadcal Overloadcal::operator*(Overloadcal n)
{
        Overloadcal d;
        d.num = num * n.num;
        return d;
}
Overloadcal Overloadcal ::operator/(Overloadcal n)
{
        Overloadcal d;
        d.num = num / n.num;
        return d;
}
int main()
{
        int x, y,z;
        cout << "Enter first number: ";
        cin >> x;
        cout << "Enter secound number:" ;
        cin >> y;
        Overloadcal ob1,ob2;
        ob1.setData(x);
        ob2.setData(y);
        cout << endl << endl;
        cout << "Arthematic Operations:" << endl;
        cout << "1.Addition" << endl;
        cout << "2.Subtraction" << endl;
        cout << "3.Multiplication" << endl;
        cout << "4.Division" << endl;
        cout << "Enter Choice(1-4): ";
        cin >> z;
        cout << endl ;
        if (z==1)
        {
                Overloadcal obj = ob1 + ob2;

                cout << "Addition of numbers = " << obj.showData()<<endl;
        }
        else if (z == 2)
        {
                Overloadcal obj = ob1 - ob2;

                cout << "Subtraction of numbers = " << obj.showData() << endl;
        }
        else if (z == 3)
        {
                Overloadcal obj = ob1 * ob2;

                cout << "Multiplication of numbers = " << obj.showData()<<endl;
        }
        else if (z == 4)
        {
                Overloadcal obj = ob1 / ob2;

                cout << "Division of numbers = \n\n" << obj.showData()<<endl;
        }


}
```

Output (Screenshot)

```
Microsoft Visual Studio Debu
Enter first number: 2
Enter secound number:3


Arthematic Operations:
1.Addition
2.Subtraction
3.Multiplication
4.Division
Enter Choice(1-4): 1

Addition of numbers = 5

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To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically
le when debugging stops.
Press any key to close this window . . .
```

Task 03

Write a program to concatenate (join) two strings by using overloaded + operator.

Code

```
#include<iostream>
using namespace std;
class Str
{
private:
	string a;
public:
	Str();
	void setStr(string);
	Str operator+(Str);
	string getStr();
};
Str::Str() : a("") {}
void Str::setStr(string n)
{
	a = n;
}
string Str::getStr()
{
	return a;
}
Str Str::operator+(Str temp1)
{
	Str temp2;
	temp2.a = (a + " " + temp1.a);
	return temp2;
}
int main()
{
	string a,b;
	cout << "Enter the first string: ";
	cin >> a;
	cout << "Enter the second string: ";
	cin >> b;
```

```
        cout << endl;
        cout << "\nInitial strings: " << endl;
        cout << "String: " << a << endl;
        cout << "String: " << b << endl << endl;
        Str s1, s2 , S;
        s1.setStr(a);
        s2.setStr(b);
        S = s1 + s2;
        cout << "\nAfter concetenation: " << endl;
        cout << "String: "<<S.getStr()<<endl;

}
```

Output (Screenshot)

```
Microsoft Visual Studio Debu

Enter the first string: The
Enter the second string: Best


Initial strings:
String: The
String: Best


After concetenation:
String: The Best

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To automatically close the console when debugging stops, enable Tools->Options->Debugging
le when debugging stops.
Press any key to close this window . . .
```